TechWhiz@LT-EDI Transformer Models to Detect Levels of Depression from Social Media Text

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01 Introduction

- Depression is a serious mental illness that profoundly impacts emotions, thoughts, and behaviors.
- Social media-related factors such as social comparison and cyberbullying contribute significantly to depressive feelings.
- Detecting levels of depression is a challenging task, as it depends on individuals' changing mindsets, which can vary over time.

02 Objective

- To predict the levels of depression
- To develop a depression detection model
- To utilize transformer models

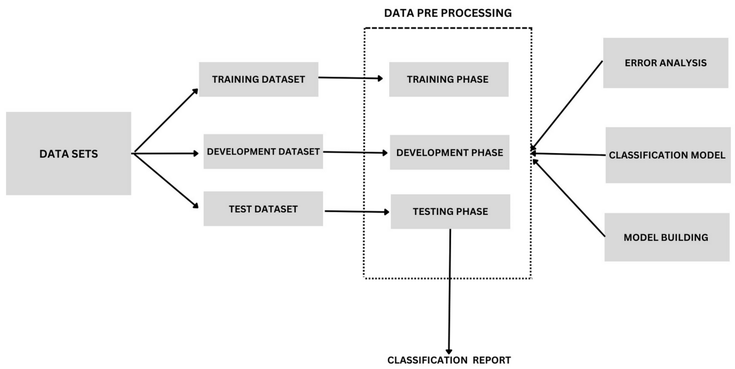
03 Dataset Description

- Dataset provided by task organizers for depression detection, consisting of training, evaluation, and test datasets.
- The training and evaluation datasets had an imbalanced class distribution, with categories for moderate, not depressed, and severe cases.
- The test dataset contained 499 instances, and the study aimed to make predictions using their proposed model for depression detection.

Category	Training dataset	Evaluation dataset
Moderate	3700	2169
Not depression	2755	848
Severe	768	316

04 Methodology

- The labels were converted into integer categorical values to facilitate input to transformer models.
- The absence of a standard structure and the presence of symbols, tags, and URLs in social media text, preprocessing was necessary.
- The proposed approach utilizes deep learning techniques to detect depression from social media text.
- The system employs three distinct Transformer models: ALBERT (base-v1), and RoBERTa (base).
- For training, 1 epoch was used for RoBERTa and ALBERT.
- The project architecture is given below:



05 Inference

- The ALBERT model's F1 score in the task reveals an imbalance-related issue with increased false positives and negatives, underlining the importance of data augmentation.
- Misclassified examples include texts with depression markers and sarcasm affecting classification.
- The metric values achieved by the two different models when evaluated on the test dataset provided for the shared task.

Model	F1-Score	Accuracy
ALBERT	0.258	0.421
ROBERTA	0.143	0.263

06 Conclusion

- The study utilized a dataset for depression detection, classifying social media text into three categories: severe, moderate, and not depressed, using models like ALBERT and RoBERTa.
- These models effectively captured linguistic cues and contextual information, aiding in depressive symptom detection from text.
- Future enhancements could focus on handling contextual information more effectively and exploring hybrid approaches involving multiple deep learning models for improved depression detection.

Reference

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